

UEN003: ENVIRONMENTAL SCIENCES

L	T	P	Cr
2	0	0	2.0

Course Objectives: The exposure to this course would facilitate the students in understanding the terms, definition, and scope of environmental and energy issues pertaining to the current global scenario. Understanding the values of regional and global natural and energy resources. Emphasize on the need for conservation of energy and environment

Definition and Scope: Importance, Public awareness and education.

Natural Resources: Introduction, Renewable and non-renewable, Forest, water, mineral, food, energy and land resources, Individual and conservation of resources, Equitable use of resources.

Ecosystems: Concept, Structure, Function, Energy flow, Ecological succession.

Biodiversity: Genetic, Species and ecological diversity, Biodiversity at global, national and local levels, Threats to biodiversity, Endangered and endemic species of India, Conservation of biodiversity.

Pollution: Definition, Pollution case studies, Causes, effects and control measures of the pollution – Air, Water, Noise, Thermal Pollution, Solid waste management, Disaster management.

Social Issues (through online platforms): Sustainable development, Environmental ethics, Climatic change, Environmental protection acts and issues.

Course learning outcome (CLO)/ Course Objectives (COs):

Upon completion of this course, the students will be able to:

1. Comprehend the interdisciplinary context with reference to the environmental issues and case studies
2. Assess the impact of anthropogenic activities on the various elements of environment and apply suitable techniques to mitigate their impact
3. Conceptualize and explain the structural and functional features of ecological systems.

Text Books:

1. Bharucha, E., Textbook of Environmental Studies, Universities Press (2005).
2. Chapman, J.L. and Reiss, M.J., Ecology- Principles and Application, Cambridge University Press (LPE) (1999).
3. Joseph, B., Environmental Studies, Tata McGraw-Hill (2006).

Reference Books:

1. Miller, G.T., Environmental Science- Working with Earth, Thomson (2006).
2. Wright, R.T., Environmental Science-Towards a sustainable Future, Prentice Hall (2008) 9th ed.